Digital Twins for Cyber-Resilient Power Grids: A Structured Corpus Review of AI Use, Threat Engagement, and Measurement Limits

Author One[a], Author Two[a]

[a]*Department of Electrical and Electronic Engineering, University Name, Dhaka, Bangladesh*

Abstract

Digital twins are becoming a reference architecture for modern power grids, with machine learning inside them. We ask whether literature naming a grid digital twin and a security term engages AI-enabled threats—and the twin itself—as an attack target, or stops at defensive use. The hypothesis came from early reading of this corpus; in its original form—defensive use common, attack-side engagement rare—it was not supported, and the framing was revised post hoc to a question of depth. Twelve OpenAlex queries yielded 1,287 records (492 eligible); a declared non-probabilistic ranking retained 100, 83 coded at full text across 46 fields. Every figure describes this corpus, not the wider literature. Machine learning is used in 70 of 83—detection in 55; 35 (42%) engage an AI-enabled threat at any level, 11 (13%) substantively; in the 29-study security-primary detection-ML core, 16 (55%) engage any, 6 (21%) substantively. Retrieval ran on defensive vocabulary, so these rates describe the defensively-framed literature only. The defensible conclusion: any-level engagement is common, depth is not reliably measured here—"any" can mean under two sentences, the substantive tier sits on the weakest coder agreement, and its threshold partly proxies document length. As coded, no grid study treats the twin itself as the adversary's objective; two read nuclear-domain records (R91, R92) attack twin interfaces and ML models, bounding it from outside; 56 of 83 name classical threats on the twin's data path—surfaces to protect, not studied targets. A five-item agenda ties each item to its gap, disconfirming evidence, and dual-use obligations.

Keywords: digital twin, power grid, cyber-physical security, adversarial machine learning, structured review, research agenda

1. Introduction

Electric power grids are digitising faster than they are being secured. Wide-area measurement, distributed energy resources, electric-vehicle charging and market-coupled control have turned the grid into a cyber-physical system whose behaviour is increasingly mediated by software, and real incidents—the 2015 and 2016 attacks on the Ukrainian distribution grid among them—have shown what software-mediated disruption costs [1, 2]. The digital twin—a synchronised, bidirectional virtual counterpart of a physical asset, updated from live sensor streams—has moved in the same decade from concept to a reference architecture for monitoring, protection and control, in the grid as elsewhere [3–5]. That same synchronisation is what makes the twin valuable and what makes it a target.

Artificial intelligence (AI) has followed the twin into the grid. Machine learning now performs state estimation, anomaly detection, load and attack forecasting, and increasingly control. The security community has been quick to adopt it. The question this review asks is whether the same community has been as quick to model AI—and the twin itself—as something an adversary can attack. Throughout, 'digital twin' must be named by the authors, so adversarial-ML work on grid security that does not mention a twin is out of scope and is not represented in any figure here.[1]

Early reading suggested a specific, testable asymmetry: grid digital-twin security research uses AI as a defensive tool far more often than it engages AI, or the twin, as an attack target. The hypothesis was formed from part of the same corpus, before coding began and with no pre-specified confirmation threshold; the analysis is therefore exploratory. And stated plainly: in its original form—defensive use common, attack-side engagement rare—the hypothesis was *not supported*. Any-level engagement is majority in the core subsets (Section 5.5); what is minority is substantive engagement. The framing was revised after seeing the data to a question of depth; the title names the measurement actually delivered rather than the asymmetry originally

[1] Citation scheme: the 100 corpus records are cited in the running text as R<ID> (R18 = Zaidi & Khan), listed in Appendix A with years, venues, venue types and citation counts, and released as bibliography keys `ref<ID>` in the reference list. Bracketed numeric citations [n] refer exclusively to non-corpus background literature.

hypothesised; and every figure below should be read with that post-hoc revision in mind. Two further limits on what the measurement can show should be stated at the outset. First, "uses ML for detection" and "engages an AI-enabled threat" are different constructs, so there is no normative baseline for what the engagement share *should* be; the measurement is descriptive, and we make no claim that a particular share would be adequate or deficient. Second, the claim the measurement supports is one of *emphasis and depth*, not of *rate*: we make no temporal "adopts faster" claim, because the corpus is a ranked, non-random subset in which a year-wise comparison is confounded by selection (Section 4). This review quantifies what the corpus does engage, and turns the measured gaps into a research agenda. It is an exploratory structured review, not an empirical deployment study.

1.1. Positioning against prior reviews

The closest work, and the reason this review had to be repositioned, is Habeeb, Weli and Almaliki's 2026 PRISMA review of "Dual-Role Digital Twins" in cyber-physical security [6]. Their contribution is real and prior: they name the dual role—the twin as both an expanded attack surface and a defence enabler—and set out four research directions around it. Their corpus is 20 studies, gathered with a query that has no specific attack operator, and spread across manufacturing, healthcare, IIoT, smart city and water management, with the power grid represented in roughly one study.

We therefore do not claim to have discovered the dual-role tension. Our position is narrower and complementary. Habeeb et al. name two axes; we measure a narrow, grid-side instance of the attack-surface axis—the twin as the adversary's explicit objective—with a published corpus, a closed threat vocabulary, and reported coder reliability. We additionally report looser tiers (the twin as an attacked data path or model, and the twin as a protected asset), because the strict-objective tier alone cannot speak to the broader attack-surface axis. In other words, they build the theory; we report a lower bound on its narrowest grid manifestation, not a test of the full axis. Section 3 sets out the mapping in full, including the five attack vectors their review identifies and our corpus's coverage of them.

1.2. Contributions

1. A screened corpus of 100 studies on digital twins for power-grid cybersecurity, drawn from 1,287 unique records of twelve OpenAlex core

queries by a *declared, non-probabilistic* top-100 ranking (relevance, citations, recency). The corpus is a ranked subset of 492 eligible records, not a census or a random sample; the ranking and its biases are reported with the corpus (Section 4).

2. A 46-field extraction of that corpus, coded by computer-assisted human reading, with the extraction method disclosed rather than hidden (Section 4).
3. A two-coder reliability study on a 20-record blind sample (24% of the 83 full-text records), reported field by field with κ, bootstrap confidence intervals, PABAK and bias indices, including the fields where agreement is only fair (Section 4.8).
4. The headline measurement, with matched denominators: machine learning is used in 70/83 full-text studies (detection in 55); 35/83 engage any AI-enabled threat and 11/83 engage one substantively; within the 29 security-primary, non-review, detection-ML studies the figures are 16/29 and 6/29. No grid study treats the twin as the adversary's objective; 56/83 name an interface or model threat in a defender frame (Section 5.5).
5. A vocabulary-level mapping from our threat terms to the five attack vectors of Habeeb et al. [6], released as a repository artifact rather than presented as evidence: it is a single-author judgment and was not double-coded (Section 3).
6. A five-item research agenda in which every item is tied to the gap that justifies it, the evidence that would validate it, and the evidence that would falsify it (Section 7).

1.3. Research questions

RQ1 How is the digital twin positioned in grid cybersecurity research—as a defensive tool, an attack target, or both?

RQ2 Which threats are named, at which grid domains and lifecycle stages, and with what threat models?

RQ3 How far does the corpus engage AI-enabled threats (adversarial machine learning, evasion, poisoning, LLM-specific, agentic)?

RQ4 How rigorous is the evidence base—evaluation method, maturity, baselines, metrics, and released code or data?

RQ5 What are the resulting gaps, and what would close them?

1.4. Structure

Section 2 gives the technical background. Section 3 reviews prior work and positions this review against it. Section 4 is the protocol. Section 5 reports results, figure-led, across RQ1–RQ4. Section 6 discusses them. Section 7 sets the research agenda for RQ5. Section 8 is the threats-to-validity analysis, and Section 9 concludes.

2. Background

2.1. Digital twins in power systems

A digital twin [3–5] is a virtual representation of a physical entity that is connected to it by a bidirectional data link, such that the virtual state tracks the physical state and, in the stronger formulations, the physical entity can be actuated through the twin. Three properties distinguish a twin from an ordinary model or simulation: synchronisation with live data, bidirectionality, and asset specificity. Grid applications span state estimation and observability, protection and control, fault and attack diagnosis, asset-health monitoring, and training and planning (R16, R24, R50).

The maturity of grid twins varies widely. Some reported twins are conceptual architectures with no implementation; many are prototypes demonstrated on simulation or testbed data; a small number report validation against real grid assets (R54, R74). This maturity gradient matters for security, because a threat model that is plausible on a conceptual twin may not be reachable on a real deployment, and vice versa.

2.2. Cyber threats to grids and to twins

The classical grid threat model is adversarial and studied: false data injection against state estimation [2, 7], replay, denial of service, man-in-the-middle on telemetry links, false command injection, and supply-chain compromise. These are conventional cyber-physical attacks, are well represented in the corpus (R1, R3), and need no AI to execute (Section 5.4).

The twinning layer adds attack surface that the classical model does not cover. The bidirectional link can be exploited in both directions; the twin synchronises with sensor streams, so any data-poisoning channel into those streams reaches the twin; and where the twin uses learned components—detectors, forecasters, controllers—those components can themselves be attacked by distribution-shift, evasion, or training-time poisoning. This is the attack-surface axis that Habeeb et al. name [6].

2.3. AI on both sides of the boundary

AI enters grid security from two directions. Defensively, machine learning performs intrusion and anomaly detection, attack classification, load and attack forecasting, and increasingly control. Offensively, the same model families are themselves vulnerable: adversarial examples can evade learned detectors [8, 9], poisoning can corrupt training data drawn from compromised sensor streams [10], and large language models introduce prompt-level and social-engineering vectors that did not exist in earlier grids. An "AI-enabled threat" in this review means an attack in which the adversary uses AI, or attacks an AI component, or both—as distinct from a conventional cyber-physical attack that happens to be detected by an AI method. These two directions have different implications (an AI-wielded attack changes the adversary; an attack on an AI component changes the victim), and the headline construct *combines* them. The coding does not separate the two directions per record; Section 5.5 reports a notes-based separation for the substantive studies and flags the lumping as a construct limitation (Section 8).

A brief terminology note. 'Digital twin' has largely superseded 'CPS model' and 'HIL simulation' in recent grids literature, so papers that functionally twin a grid without using the phrase are out of scope of the search, despite carrying relevant threat models.

3. Related work and positioning

3.1. The closest review

Habeeb, Weli and Almaliki (2026, *Digital Twin*) review Dual-Role Digital Twins for cyber-physical security with PRISMA 2020: 412 identified, 318 screened, 72 eligibility-checked, 20 included, spanning January 2019 to December 2025 [6]. They ground the dual role in Complex Adaptive Systems, functional duality, and zero-trust architecture, and they name five attack vectors. Their search string is `("Digital Twin" OR "DT") AND ("Cybersecurity" OR "Attack Surface" OR "Defense" OR "Honeypot" OR "Intrusion Detection")`. It contains no term for a specific attack (no adversarial machine learning, poisoning, evasion, or reverse-pathway operator), so attack-side records have to enter through the security and defence terms. The same limitation applies to our twelve core queries, which is why we added the four adversarial-AI probes.

3.2. The bridge: their vectors, our vocabulary

Our codebook's threat vocabulary was designed before we read their review, according to the dated codebook and design log in the repository—a claim that cannot be verified from this text alone. The review's *positioning*, by contrast, was revised after reading their work (Section 1.1): the framing changed, the vocabulary did not. We state both facts separately and withdraw any stronger independence claim than that. A vocabulary-level mapping from our threat terms onto their five attack vectors exists—every one of their vectors is expressible in our closed vocabulary, and our vocabulary's LLM-specific (4 records) and agentic (2) constructs fall outside their five vectors altogether—but the mapping is a single-author judgment that was not double-coded, so it is released as a repository artifact (`VECTOR_MAPPING.md` in the repository's `analysis/` directory) and is not presented in this paper as evidence that any vector is dealt with empirically.

The relationship between the two search designs is the mirror image of the mapping. Our search adds retrieval operators their query does not contain—adversarial machine learning, poisoning, evasion and large-language-model terms—so records whose visibility depends on those terms could enter our pool but not theirs; the added terms that do not constitute novel constructs map onto their own vectors, which is why the mapping claims coverage rather than novelty, and why the classical `dos` and `fdi` terms are placed there under the vector whose surface they exploit rather than by an identity of mechanism.

3.3. Other reviews in the corpus

Several corpus entries are themselves reviews (R1, R3, R16, R24, R50, R57). Their coverage is broad on the defensive side. Our contribution relative to them is the quantified attack-side measurement: we do not merely note that AI threats matter, we count how often the corpus engages them and find that any-level engagement is common—majority in the core subsets—while substantive engagement is the minority case (11 of 83 full-text records). The finding is one of depth, not of presence versus absence.

4. Methodology

4.1. Design

This is a *structured* literature review reported in PRISMA-2020 flow format for transparency of search and selection. It is not a systematic review in

the PRISMA sense: it searches one database (OpenAlex), has no registered protocol, and retains a ranked, non-random subset of the eligible records. We use PRISMA flow reporting because the alternatives (a narrative review) would hide the selection rule rather than expose it. The protocol was fixed before extraction, with the codebook revisions noted below. The review is theory-testing in intent: the asymmetry stated in the introduction is the hypothesis, and the retained 100 studies are the corpus on which it is explored—with no pre-specified confirmation threshold, as the introduction states.

4.2. Search strategy

We searched OpenAlex with twelve core queries combining a digital-twin term, a grid-domain term, and a security term, yielding 1,287 unique records after de-duplication. We then ran four additional adversarial-AI probes—combining digital-twin and grid terms with adversarial machine learning, poisoning, evasion, and large-language-model terms—which raised the candidate pool to approximately 2,000. The probes exist because the core grid-security vocabulary is not expected to retrieve the attack-side literature.

What the released record does *not* establish is the probes' effect on the corpus. The screened set of 1,287 records derives from the twelve core queries; per-record query provenance for the probe-added records (roughly 700) was not retained, so we cannot report how many of the 492 eligible, or of the retained 100, are retrievable only through the probes. The retrieval-bias defence that the probes were designed to provide therefore cannot be made from these data, and we do not make it (Section 8, Search bias). Releasing the query strings, the run dates and the per-record provenance is a precondition for resolving this. Two artifacts now exist in the repository: a provenance manifest (‘SEARCH_PROVENANCE.md‘) stating exactly what must be released, and a dated snapshot (‘analysis/search_snapshot_2026-10-19/‘) containing *reconstructed* query strings (the originals were not preserved) together with a second-database (Crossref) run of each string on 2026-10-19. The original strings themselves remain a required artifact, and OpenAlex reruns attempted on 2026-10-19—twice, the second during this revision—were blocked both times by the API's shared-IP budget limit. What this means for the results is stated here rather than only in Section 8: the corpus derives from defensive-vocabulary retrieval, so any attack-side rate measured on it—including the RQ3 headline—is conditional on that retrieval design

and says nothing about the attack-side literature the queries do not retrieve (Section 8, Search bias).

4.3. Selection and inclusion

We screened the 1,287 core-query records on title and abstract. A record was eligible if its title or abstract contained at least one digital-twin term, at least one grid, power or energy term, and at least one cyber or security term; 795 records failed at least one of the three conditions, leaving 492 eligible.

The grid term is deliberately broad—*grid, power or energy*—and the eligibility rule therefore admits adjacent domains. Four retained records are outside the power-grid domain on a strict reading (a vehicle-twin threat-modelling review, a cyber-fraud study, a network-traffic anomaly-detection paper with no grid or twin instantiation, and a nuclear physical-protection risk analysis in which no twin is instantiated); three more are borderline (nuclear digital twins, battery twins for telecom supply, and a security-education cyber twin). A strict-scope sensitivity analysis that removes these records is reported alongside every headline figure (Section 5.5); the borderline records are listed there with their identifiers.

The eligible set was then narrowed to 100 by a non-probabilistic rule combining OpenAlex relevance score, citation count, and recency, with the final cut at rank 100. This ranking is a *selection* step, not an eligibility criterion. Its components pull in opposite directions—citations favour older, established work; recency favours new work—but which component dominated the realised cut cannot be established from the released record, and the corpus's ID ordering does not speak to it. The IDs are best read as a display order: year descending, and within each year block citations descending (IDs 1–45 are all 2026; per-record citation counts are given in the Cit. column of Appendix A). That pattern is compatible with a composite rule followed by this sort, and with a purely year-first rule only if the 492 eligible records contained at most 76 records from 2025–26, which the released record cannot check; the ID ordering therefore carries no information about the selection rule. The clean test is the year distribution of the 492 eligible records against the retained 100, which would show directly which years the cut discarded. The 492-record export was not preserved; the provenance manifest lists its release as a required artifact, so the test cannot be run on the released record (Section 8, Search bias). What the released record does establish is the realised corpus's profile: 40 of the 100 retained records have zero citations at retrieval, 76% of the corpus is from 2025–26, and 24

of the 100 are from 2020–2024. The corpus is therefore recency-heavy; we report that profile and make no claim about which component produced it. The profile does favour reviews (36% of the retained corpus). Because the retained subset is not a random sample, every prevalence figure in Section 5 is a statement about the retained 100 studies and should not be read as an estimate of the wider literature; the corpus's 76% share of 2025–26 records is partly an artefact of the selection, not a finding about the field. Small denominators bound all percentages, and the uncertainty that a confidence interval should carry here is coding misclassification, which we quantify in Section 5.7 rather than with sampling intervals.

4.4. Screening

Screening was title-and-abstract, applying the three-term eligibility rule above: 795 of the 1,287 core-query records were excluded for missing at least one of the three terms (digital twin; grid, power or energy; cyber or security), leaving 492 eligible records. Inclusion then ranked the eligible set by relevance, citation count, and recency, and retained the top 100, as declared in the selection step above. Figure 1 is the flow. The flow reports the core-query pipeline; the approximately 700 probe-added records are shown at identification but their screening status is not established in the released record, as Section 4 discloses.

4.5. Extraction

We extracted 46 fields per record. Extraction was computer-assisted but human-coded: a script surfaced keyword hits and context snippets for each field, and every code was assigned by a human reading those snippets in context. Keyword snippets were scored against the full text and the surrounding paragraph, not the excerpt alone; reference lists and author biographies were stripped before scanning, so absence claims rest on the surviving body text. The extraction is reproducible from the coded excerpts and DOIs, which are released.

Of the 100 records, 83 were available in full text and coded in full; 17 were available only as abstracts and were coded at abstract level only. All abstract-only fields that require full text are treated as missing, not as zero, throughout. The 17% non-retrieval rate is an access limitation of the authors (paywalled or blocked full texts), not a property of the literature or a reproducibility finding; the affected records are listed with DOIs so readers can see exactly what is missing (Appendix A). The 17 are not a random subset

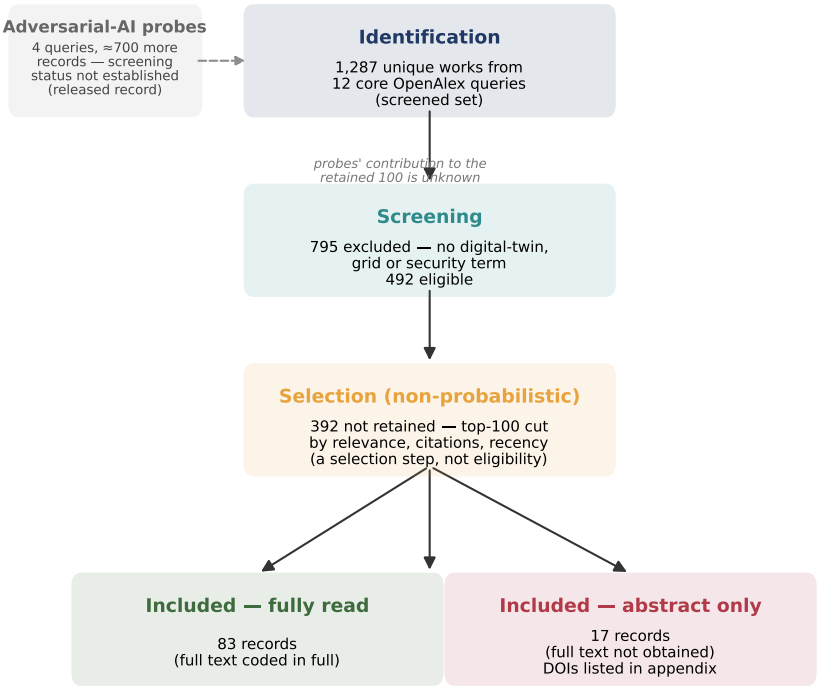


Figure 1: PRISMA-style flow. The 1,287 screened records derive from the twelve core queries; the four adversarial-AI probes added roughly 700 candidate records whose screening status is not established in the released record (Section 4). The top-100 cut is a declared non-probabilistic selection step, not an eligibility criterion. Eighty-three records are coded at full text and 17 at abstract level only, because their full text was not obtained.

of the corpus, and the loss is not neutral for the headline: an abstract-level reading of the 17 (title plus abstract) codes 15 as *tool* and the two exceptions—R91 and R92, Spirito et al., OSTI/Idaho National Laboratory reports on nuclear reactors—as attack-side work. Both were obtained and read during this revision: R91 in full text (48 pages, retrieved from the INL digital library; the OSTI full-text endpoint was unreachable from our environment) and R92 from its extended report abstract on OSTI (two full-text retrieval attempts failed). R91 analyses attack scenarios against the twin itself—an SSRF exploit of a cloud twin instance used to exfiltrate the twin and replicate it offline in simulation mode, and data deception of the twin's autonomous control ("by modifying the data received by the digital twin,

it was deceived into throttling the output from the power supplies")—while R92 constructs multi-rate hyperparameter attacks on the twin's ML models, with an event-triggered predictive-control response demonstrated in simulation. A post-hoc, single-author reading codes both as *both*-role records (attack side and countermeasures; the per-record coding is released as `R7_READBACK.md` in the repository's `analysis/` directory); neither is merged into the frozen matrix, and neither is a grid-domain record. The strict-target null below therefore carries its "grid-domain" qualifier on read evidence, not only on abstract labels: the corpus's own nuclear-domain records demonstrate twin-directed attack work (Section 5.3).

4.6. Codebook

The codebook defines each field and its controlled vocabulary, and was revised five times during extraction. The revisions are logged in the repository and are load-bearing:

- **R2** distinguishes a review from an enabling paper, and classifies attack-generating work as *bridge*.
- **R3** requires a named interface or protocol before `attack_surface_named` may be coded *yes*; passing mentions do not qualify.
- **R4** prevents passing mentions from triggering the *multi* lifecycle value.
- **R5** contracts `ml_family` to a controlled vocabulary.
- **R6** fixes evaluation-method precedence (*hardware-in-the-loop* > testbed > simulation) and maps surveys to *concept* maturity.

The `twin_role` field requires positive evidence. *both* needs explicit coverage of the defensive and attack-target roles *and* of the tension between them; *target* requires that the twin be the adversary's objective rather than an asset to be protected. *both* = 0 is therefore definitional: a paper cannot score *both* without stating the tension, and we report that zero as a statement about *explicit articulation*, not as evidence that dual-role practice is absent. The category is attainable—the two nuclear-domain records read during this revision (R91, R92) both read as *both* (Section 4 on extraction)—so the zero among the 83 grid-frame full texts is a statement about this corpus, not about the category. The strict *target* rule is likewise one end of a tiered measurement: in a security setting an asset that needs protecting is an asset an

adversary may attack, so Section 5.3 reports three tiers side by side—twin as the adversary's objective (strict), twin's data path or learned components named as an attack surface (interface and model threats from the closed vocabulary), and twin as protected asset (*tool*). Candidates such as R84, "Enhancing the cybersecurity of battery digital twins," were read as *tool* because the twin remains an asset under protection; under the looser tiers that study counts as naming an interface threat, which is the reading its content supports.

4.7. Scoring of AI-threat engagement

Each AI-enabled threat field was coded on a 0–2 ordinal scale: 0 = not addressed; 1 = mentioned, listed, or peripheral (fewer than two sentences or no failure-mode detail); 2 = substantive (a threat model, mechanism, experiment, or mitigation specific to that vector, or a section dedicated to it). The threshold was applied to the full text rather than the abstract. The threshold also makes the substantive tier partly a proxy for document length: a conference-length paper structurally has less room than a journal article or a thesis to clear "a section dedicated to it", so depth comparisons across venue formats conflate engagement with length. The corpus shows the expected signature—substantive engagement in 9 of 62 journal articles, 1 of 4 preprints and 1 of 1 thesis, but 0 of 14 conference papers (Section 5.5)—and the depth claim is qualified accordingly wherever it is made. `ai_threat_overall` is 1 if any vector is 1 or 2, and 2 if any vector is 2. Both coders coded the scale independently; per-vector agreement is reported in Section 5.7.

4.8. Reliability

A second coder independently coded 20 of the 83 full-text records (24% of the full-text corpus, drawn from the full-text frame only), blind to coder 1's values and to the screening label. Both coders are authors of this review; coder 1 is the codebook author and coded 80% of the corpus. The sample of 20 contains about 0.2 expected *target* cases at corpus prevalence, so `twin_role` agreement carries almost no information about the rare class that the headline needs; the same holds for the rarer AI vectors. Agreement is reported field by field in Section 5.7 as Cohen's κ with bootstrap 95% intervals, PABAK (prevalence-adjusted bias-adjusted kappa, which removes the prevalence depression that affects κ), and a bias index (the largest between-coder difference in category marginals), together with percentage agreement where a field is near-constant and κ is undefined. With $n = 20$, the large-sample

standard error of κ is of the order of 0.2–0.3 even for the better fields; cells in Table 7 are point estimates, and a stable ranking of field difficulty is not achievable from this sample.

4.9. Adjudication

All field-level disagreements were adjudicated against the codebook and the underlying evidence. The rules, stated here rather than buried, are: adopt the value the codebook and evidence dictate; where both coders' values are defensible, coder 1's prevails; where neither fits, the codebook rule decides. Adjudication changed 20 net cells (Section 5.7). No headline field moved, but under a tie-break that defaults to coder 1 this is the expected outcome, not evidence of robustness: headline fields could only have moved had the adjudicator overridden coder 1, and the adjudicator works with coder 1. Adjudication therefore does not test the headline; the released pre-adjudication codes are the honest artefact, and they are published frozen alongside the adjudicated matrix.

4.10. False-negative audit

Because the headline is an absence—of attack-side engagement—we audited it, and we report the audit with its design limits stated rather than implied. We ran a strict AI-threat lexicon over the 48 records coded as *not engaging* any AI threat and a broader eleven-family net. Both the extraction and the audit are keyword-assisted, so the audit shares the extraction's blind spot: "no engagement" partly means "no keyword hit", and asking a lexicon to re-find what a lexicon missed is close to circular. Its positive control recovers only 7 of the 11 substantive positives, and its recall on mentioned-only (score-1) engagement—the harder case, covering 24 of the 35 engaged records—was never measured. Of the 48 negatives, 38 fired the broad net and 23 fired a strong family; the 23 strong firings were individually inspected at audit time, and the 15 weak-only firings were inspected during this revision by an author reading the released audit windows and stored abstracts (post hoc, single-reader; Section 5.8). A seeded 12-record sample of the 48 negatives was re-read the same way. The audit is retained as a disclosure of the false-negative channel, not as a bound on it; the design that would bound it—an independent full-text re-read of a random sample of the 48 negatives by a coder outside the author team—is stated as required future work in Section 5.8.

4.11. Verification and reproducibility

Every DOI was verified against an authoritative registry: 96 of 97 resolved with a title match (95 exact, one near), one DOI is not registered, and three corpus records carry no DOI. Corpus metadata was cross-checked against Crossref, and 18 of 54 venue labels disagreed, which we resolved in favour of the registry. All extraction data, the codebook, the coding sheets, the reliability script, and the figure script are released as a reproducibility package.

5. Results

5.1. Which denominator supports which claim

Table 1 summarises the denominators behind every figure in this section. Comparisons are valid only within a matched denominator; the review's headline comparison (defensive ML use against AI-threat engagement) is reported on all of rows 4–8 so that no reading depends on which subset is chosen. Two properties of the table should be stated once here. First, the subsets are nested re-slices of the same coded data, not independent replications: that their directions agree is expected whenever the shared codes are stable, and the table is a transparency device, not a robustness battery. Second, one concentration is structural rather than analytic: all 16 any-engagers among the 41 security-primary, non-review studies fall inside the 29-study detection-ML core, so the core subset is where engagement lives by construction. The core was defined during analysis—it is the subset with the strongest obligation to model attacks on its own detectors—not pre-registered, and its majority-any reading carries that provenance with it.

5.2. Corpus profile (RQ1, descriptive)

The corpus is 100 studies: 67 journal articles, 17 conference papers, 5 theses, 4 preprints, 3 reports, and 4 book chapters. Publication is recent and accelerating: 1 study from 2020, 5 from 2021, 2 from 2022, 8 from 2023, 8 from 2024, 31 from 2025, and 45 from 2026; 76% of the corpus appeared in 2025 or 2026 (Figure 2a). That concentration is partly a selection effect of the ranked, non-random cut (Section 4), as Figure 2's caption states, and must not be read as a finding about the literature's growth. The leading publishers are MDPI (17), Elsevier (12) and IEEE (8). Table 2 gives the profile.

Table 1: Denominators used in this review (full-text records unless stated).

Subset	n
Retained corpus (all records)	100
Full-text records	83
Abstract-only records	17
ML-used studies	70
ML-for-detection studies	55
Non-review studies	49
Security-primary, non-review studies	41
Security-primary, non-review, detection-ML	29
Strict-grid-scope records (Section 5.5)	79
Journal articles (full text)	62
Peer-review-filtered journal subset (Section 5.6)	58

5.3. How the twin is positioned (RQ1)

Across all 100 records, the adjudicated stance distribution is defensive 42, review 36, bridge 10, enabling 10, adversarial 2; security is the primary subject in 83 of 100 (Figure 3a). The distribution differs from the screening seed—which had adversarial 11, review 23—confirming that keyword-level stance labelling is unreliable and that reading is necessary (Figure 3b).

The sharper result is `twin_role`, coded on the 83 full-text records: the twin is treated as a defensive *tool* in 75—a near-constant default (90%) on which coder agreement therefore carries almost no information about the rare classes—as an attack *target* in exactly 1, as *both* in 0, and is *unclear* in 7. The single strict *target* record (R59) is a vehicle-twin threat-modelling review, outside the grid domain; under a strict grid scope the target count is zero. The *both* = 0 result is definitional (the codebook requires explicit articulation of the tension), so it records the absence of *explicit articulation* of the dual role that Habeeb et al. name [6], not the absence of dual-role practice.

Because "asset under protection" and "adversary's objective" are not separable in a security setting, Table 3 reports the tiered measurement side by side. Beyond the single strict-target record, 56 of 83 full-text studies name an interface threat on the twin's data path (man-in-the-middle 26, replay 23, reverse pathway 10, false command injection 22; interface threats combined 44) or a model threat on its learned components (poisoning, evasion,

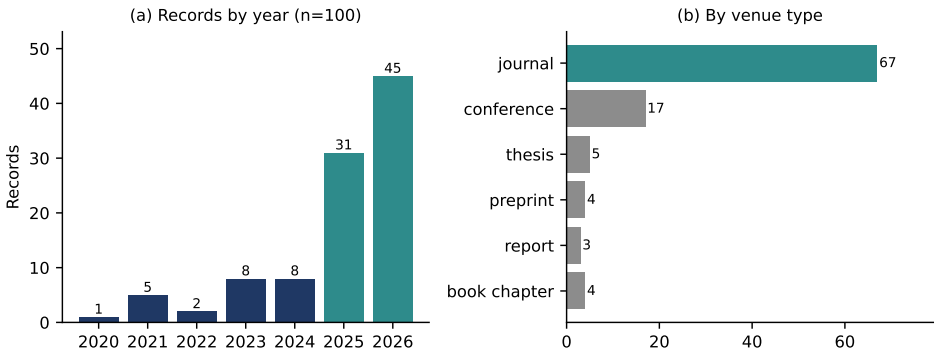


Figure 2: Corpus profile. (a) Records by year. The 2025–2026 concentration reflects the non-random top-100 selection as much as literature growth (Section 4). (b) Records by venue type.

prompt-level; 27, overlapping the interface set in 15 records). Of the 75 *tool* records, 51 name such a threat: the dominant pattern in the corpus is a twin defended against threats whose target surface includes the twin itself, without the twin being treated as the adversary's objective. The strict rule is not blind to contrary language: the corpus contains one-sentence assertions that the twin "is a main target for hackers" (R48) and that a compromised twin could cause "unauthorized data access, manipulated simulations" (R88); both were read and set aside as undeveloped, non-AI statements, which is the strict rule working as designed rather than evidence suppression. Two qualifications keep the middle tier honest. First, the interface threats are classical grid-communication attacks that any twin-based defender would name; the tier measures defender-framed threat-*naming* that bears on the twin's data path, not that the twin's own surface is the analytical object of the study, and the tier label is chosen accordingly. Second, the most direct field for this claim, `attack_surface_named`, has prevalence 55/83 (66.3%) and the lowest agreement in the reliability study on both statistics ($\kappa = 0.14$, PABAK 0.20, AC1 0.31; Section 5.7), so the tier boundaries carry that uncertainty.

This is the first half of the asymmetry, stated precisely and read as a descriptive observation of the coding rather than a finding about the field: *no grid study in this corpus treats the twin as the adversary's objective*, while the twin's data path and learned components are named as attack surfaces—to be defended—in roughly two thirds of the corpus. The two single-record exceptions that keep the strict and design-phase columns non-empty (R59,

Table 2: Corpus profile ($n = 100$; full text = 83, abstract-only = 17).

Characteristic	Records
Journal article	67
Conference paper	17
Thesis	5
Preprint	4
Report	3
Book chapter	4
2026	45
2025	31
2024	8
2023	8
2022	2
2021	5
2020	1
Stance: defensive	42
Stance: review	36
Stance: bridge	10
Stance: enabling	10
Stance: adversarial	2

R97) are both outside the grid domain, so both headline zeros rest on out-of-scope records rather than on in-domain counterexamples; and the two read nuclear-domain records (R91, R92; Section 4) demonstrate twin-directed attack work—against the twin's interfaces and its ML models—that the grid-domain coding does not contain.

5.4. Threat vocabulary and lifecycle concentration (RQ2)

Table 4 reports the closed threat vocabulary. Denial of service (49), false data injection (46) and malware (34) lead; the AI-relevant terms that the corpus does contain— poisoning (22), evasion (11)—are present but well behind the classical attacks. A mean of 3.77 distinct threats is named per full-text study (median 3, max 11), and 10 of 83 studies name no threat at all, which is itself notable in a security review. These counts sit on the least reliable data in the review: in the double-coded sample, `threat_models_named` exact-set agreement is 5% (mean Jaccard 0.43) and `threat_count`

Table 3: Tiered measurement of the twin's position relative to the attack (full text, $n = 83$). The middle tier means the record names at least one interface threat (`mitm`, `replay`, `reverse_pathway`, `false_command`; 44 records) or model threat (`poisoning`, `evasion`, `prompt`; 27 records; 15 records name both kinds). The interface threats are classical grid-communication attacks, so the tier records defender-framed threat-naming that bears on the twin's data path, not that the twin is the study's analytical object. *tool* is the coders' near-constant default (75 of 83). The tiers are descriptive observations of this coding, not findings about the field.

Tier	Records
Twin as adversary's objective (strict `target`)	1
of which grid-domain	0
Interface or model threat named, defender-framed	56
Twin as protected asset (`tool`; near-constant default)	75
of which also name an interface or model threat	51
Twin position unclear	7
`attack_surface_named` = yes ($\kappa = 0.14$)	55

differs by a mean of 1.8 threats against a mean of 3.77 (Section 5.7). The vocabulary counts should therefore be read as indicative of scale and ordering, not as exact prevalences.

Threat coverage is concentrated almost entirely at the operation phase. Figure 4 crosses the threat vocabulary with `lifecycle_stage`. The design and commissioning columns are effectively empty—one design-phase study (R97), itself outside the grid domain, and none at commissioning—while the operation column is dense. Three features of the figure need saying. First, the single design-phase record R97 names only an out-of-vocabulary threat (coded `other`), so no vocabulary row registers in the design column; the column is empty because of what the one design study names, not because it names nothing. Second, that record is the nuclear physical-protection risk analysis outside the power-grid domain (R97), so within a strict grid scope no record at all is coded at design. Third, the maintenance and decommissioning columns are empty as well, and no record is coded at either stage: the codebook permits both values, so the emptiness is a finding of the coding, on a field of moderate agreement ($\kappa = 0.32$, PABAK 0.59, AC1 0.54—not the study's weakest field; that is `attack_surface_named`, $\kappa = 0.14$, AC1 0.31). Across the corpus, lifecycle is operation in 66 of 83 studies (79.5%), *multi* in 12, unclear in 4, design in 1, and commissioning in 0, as coded. Read as

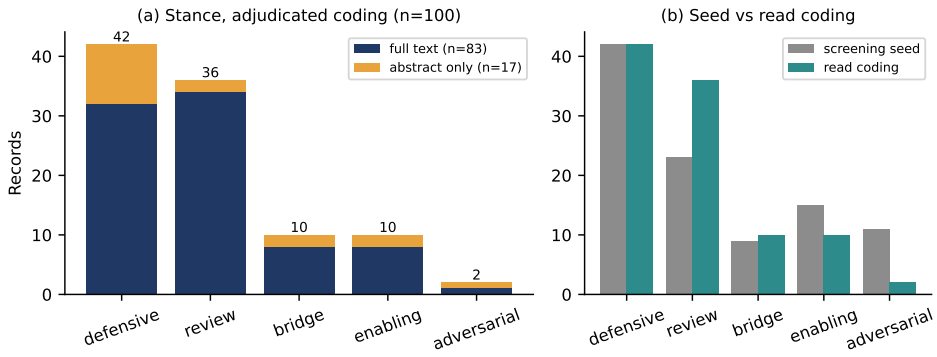


Figure 3: Stance. (a) Adjudicated coding across all 100 records, split by whether full text was coded. (b) Screening seed versus read coding: the shift from *adversarial* to *review*/*defensive* shows the seed over-called an adversarial stance on keyword evidence.

single labels, security work concentrates where the twin runs at deployment, not where it is built.

Threat models are also thin. An explicit, stated threat model appears in 19 of 83 studies, and an adaptive adversary is modelled in 10—so the corpus mostly names threats without committing to an attacker model, which is precisely the weakness that makes attack-side claims hard to test.

5.5. Engagement with AI-enabled threats (RQ3)

Note on the two threat field families, unified here for the rest of the paper: the closed classical vocabulary (`threat_models_named`) and the AI-engagement fields (`ai_*`) overlap in vocabulary but differ in construct—naming a threat versus engaging it on the 0–2 scale. The overlap is near-nested in the data: of the 22 records naming `poisoning`, 19 are coded `ai_poisoning` ≥ 1; of the 11 naming `evasion`, 9 are coded `ai_evasion` ≥ 1; `prompt` is named once while `ai_llm` ≥ 1 in 4 records. The `ai_*` fields are the operative measurement for AI-threat engagement; the classical vocabulary's AI-adjacent terms are retained only for family-level reading (Section 5.7), and the residual named-but-not-engaged records (poisoning: R64, R68, R87; evasion: R15, R51) are classical usages—ARP poisoning, the SolarWinds incident as a data-poisoning example, evasion of non-ML supervisors. This is the headline, reported with matched denominators. Machine learning is used in 70 of the 83 full-text studies; its dominant role is detection (55), followed by forecasting (12), none (13), and control (3). Forecasting and control

Table 4: Threat vocabulary (full text, $n = 83$). Terms are non-exclusive.

Threat	Records
Denial of service (`dos`)	49
False data injection (`fdi`)	46
Malware (`malware`)	34
Man-in-the-middle (`mitm`)	26
Replay (`replay`)	23
False command injection (`false_command`)	22
Data/model poisoning (`poisoning`)	22
Physical attack (`physical`)	21
Insider (`insider`)	20
Supply chain (`supply_chain`)	19
Model evasion (`evasion`)	11
Reverse pathway (`reverse_pathway`)	10
Other	8
Hardware	1
Prompt-level	1
No threat named at all	10
Mean distinct threats per study	3.77

are not security functions, so the security-relevant defensive figure is 55/83 (66%), not 70/83. Across the same 83 records, 35 (42.2%) engage at least one AI-enabled threat at any level, 11 (13.3%) engage one substantively, and 48 (57.8%) do not engage at all (Figure 5). All 35 engaged studies are among the 70 that use machine learning at all, and 30 of the 55 (55%) detection-ML studies engage an AI threat—a majority—so the non-engagement among ML-using studies is 35 of 70 (50%) rather than a zero/one split. The defensible reading of these numbers is the one the instrument supports: *any-level engagement is common in this corpus, and depth is not reliably measured here.* Three reasons, stated once here and detailed in Sections 5.7 and 8. (i) "Any" can mean fewer than two sentences. (ii) The substantive tier is coded on the weakest collapse in the study ($\kappa = 0.14$, with only 1 of 6 positives agreed by both coders). (iii) The substantive threshold is partly a document-length proxy—0 of 14 conference papers clear it, against 9 of 62 journal articles, 1 of 4 preprints and 1 of 1 thesis—so "depth" is confounded with venue format even within the non-review set. And one condition frames

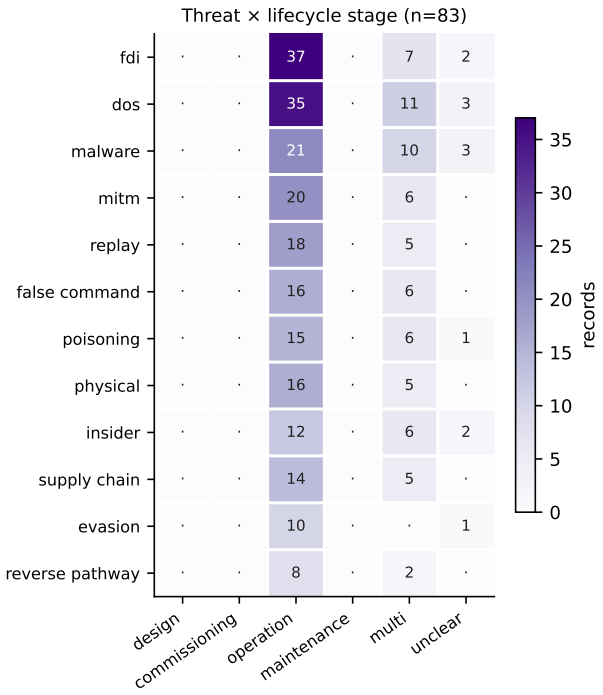


Figure 4: Threat × lifecycle-stage coverage (full text, $n = 83$). Empty cells in the design and commissioning columns are a coding outcome, not a census of practice; the field's agreement is $\kappa = 0.32$ (PABAK 0.59). The single design-phase record names only an out-of-vocabulary threat (`other`), so no vocabulary row registers in the design column. No record is coded at maintenance or decommissioning.

everything: the corpus was retrieved by defensive vocabulary (Section 4), so these rates describe the defensively-framed literature, not the attack-side literature the queries do not retrieve.

Three denominator restrictions probe it. Restricting to the 49 non-review studies: 20 of 49 (41%) engage any, 6 of 49 substantively. Restricting to the 41 security-primary, non-review studies: 16 of 41 (39%) engage any, 6 of 41 substantively. Restricting to the 29 security-primary, non-review, detection-ML studies—the subset with the strongest obligation to model attacks on their own detectors—16 of 29 (55%) engage any AI-enabled threat and 6 of 29 (21%) substantively. Majority any-level engagement therefore holds on the two core denominators only—the 29-study core subset and the 55 detection-ML studies (16/29; 30/55)—while on every wider denominator it is minority (35/83, 42%; 20/49, 41%; 16/41, 39%); substantive engagement

Table 5: Year-wise profile of full-text records ($n = 83$). ML = `ml_used` = yes; "any" = engages at least one AI-enabled threat; "subst." = substantive. Corpus years are shaped by the non-random selection rule, so these are descriptive counts, not a temporal estimate.

Year	n	ML	any	subst.
2020	1	0	0	0
2021	3	0	0	0
2022	2	2	1	1
2023	6	4	0	0
2024	7	6	1	0
2025	27	23	15	6
2026	37	35	18	4

is minority on every denominator (at or below one fifth). Majority defensive use thus coexists with majority any-level engagement in the core subsets and with minority substantive engagement everywhere; the honest summary of the asymmetry is that engagement is common but shallow in the core subsets and thin outside them, and there is no external baseline for how deep it should be. These restrictions re-slice the same coded data (Section 5.1), so their agreement shows stability of the shared codes, not independent robustness. The same pattern holds under the strict grid scope inside the core subset: removing R18 and R69 leaves 27 studies, of which 15 (56%) engage any and 5 (19%) substantively.

Because the corpus is a ranked, non-random subset, a year-wise comparison cannot support a rate claim; Table 5 reports it as description only. Within the corpus, ML use is 85% or better from 2024 onward (6 of 7 in 2024, 23 of 27 in 2025, 35 of 37 in 2026; 2023 is 4 of 6, 2022 is 2 of 2), and substantive engagement in the three years with at least five records (2024–2026) never exceeds 22.2% (6 of 27 in 2025); the small-denominator years are not interpretable as rates—2022 shows 1 of 2 (50%) substantive—and the 2020–21 and 2023 cells show zero engagement on small denominators. We draw no temporal conclusion from any of these cells.

Per vector, at any/substantive level: adversarial machine learning 24/8, data or model poisoning 19/7, evasion 9/1, LLM-specific 4/2, and agentic attack automation 2/1.

The construct behind these counts combines two directions that should be reported separately: attacks *on* AI components and attacks *wielded with* AI. The coding does not separate them per record. A notes-based reading

of the eleven substantive studies classifies four as attack generation with AI (adversarial-ML or GAN synthesis of attacks; R22, R49, R57 and R69—the last of unattributed authorship in a low-tier venue and outside the strict grid scope, flagged again in Section 8), five as attacks on learned components (poisoning, evasion or prompt-level risks; R16, R17, R24, R54, R74), and two as both (R50, R94); the any-engagement count of 35 mixes the two directions in unknown proportions. The headline is therefore an engagement figure, not an attack-type figure.

A strict-scope sensitivity addresses the breadth of the grid term. Excluding the four records outside the power-grid domain on a strict reading (R18, R59, R69, R97: a network-traffic anomaly-detection paper with no grid or twin instantiation, the vehicle-twin review, the cyber-fraud study, and the nuclear physical-protection risk analysis) leaves 79 full-text records: 33 (41.8%) engage any AI-enabled threat, 10 (12.7%) substantively, ML use 67, detection 52, and the strict twin-as-target count is zero. Excluding additionally the three borderline records (R34, nuclear digital twins; R84, battery twins for telecom supply; R68, a security-education cyber twin) leaves 76 records: 32 (42.1%) any, 10 (13.2%) substantive. Within the 29-study core subset the same strict scope removes R18 and R69, leaving 27 studies: 15 (55.6%) any, 5 (18.5%) substantive (and 25 studies, 14 any and 5 substantive, if the borderline records are also removed). The asymmetry's direction is robust to scope; its magnitude moves by at most about one and a half points on the any-level share.

Two patterns sharpen the result. First, engagement is shallow: mentioned (score 1) is more common than substantive (score 2) for every engaged vector except LLM threats (2 mentioned, 2 substantive, a tie on a base of 4) and agentic threats (1 mentioned, 1 substantive, a tie on a base of 2). Second, engagement is concentrated: every substantive evasion case, every substantive agentic case, and one of the two substantive LLM cases sit in a single review (R50), with the other substantive LLM case in R16 (at any level, 3 of the 4 LLM-engaged records are reviews). On the review authorship question the substantive set has 5 reviews (45%) against 34 (41%) in the full-text base; both counts are small and the difference is unsupported by any test.

Table 6 lists the eleven substantive studies. Nine are journal articles (all 2025–26), one is a thesis (R74) and one a preprint (R94, 2022); five carry the *review* stance and three the *bridge* stance, so 8 of 11 are review-or-bridge work. All eleven use ML predominantly for detection. No study's primary contribution is an AI-enabled attack on a grid digital twin; the

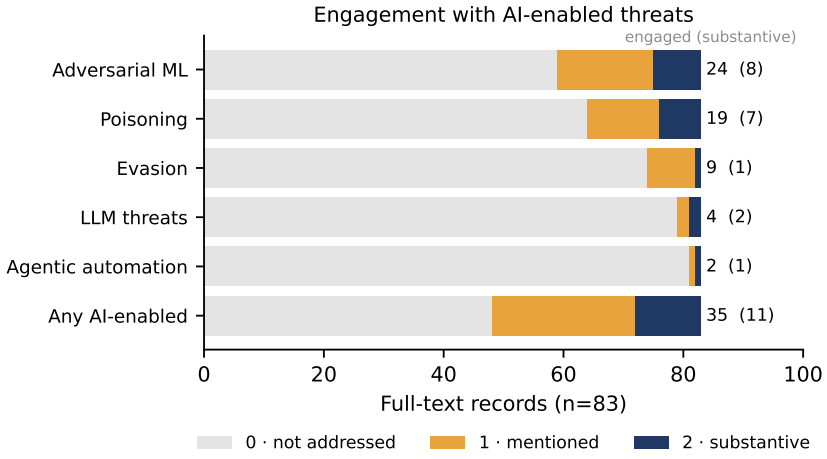


Figure 5: Engagement with AI-enabled threats (full text, $n = 83$). Scale: 0 not addressed, 1 mentioned or peripheral, 2 substantive. The bar labels give engaged (substantive) counts.

deepest engagements are themselves reviews (R16, R50). One of the eleven (R69) lies outside the grid domain on a strict reading, is excluded from the strict-scope counts above, and is examined further in Section 8.

Two quality-bound sensitivities qualify the substantive count. Excluding R69 alone (unattributed authorship, IRJMETS venue, outside the strict grid scope) leaves 10 of 82 substantive at the full-text denominator and 5 of 28 in the core subset; the peer-review venue filter (Section 5.6) leaves 8 of 58. A citation-based split runs the other way: among the 50 cited records, 18 (36%) engage any and 5 (10%) substantively; among the 33 zero-citation records, 17 (52%) and 6 (18%). The uncited, mostly 2025–26 records engage *more*, so shallow engagement is not an artefact of weak records alone—though the zero-citation subset's higher engagement is itself consistent with the corpus's recency- and review-heavy profile, and no available filter cleanly separates paper quality from field behaviour (Section 8).

Read qualitatively, the eleven cluster into four patterns. (1) Generative and adversarial modelling as attack generation: the twin is used to synthesise adversarially consistent attacks and GAN-based evasion by the authors themselves (R16, R24, R49, R50, R57, R94). (2) Learned intrusion detectors stressed under distribution shift: data poisoning (R17, R74, R94), FGSM/JSMA perturbations (R50), and GAN-driven evasion (R49, R94). (3) LLM threats treated as architecture-level hazards—hallucination, prompt in-

Table 6: The eleven studies engaging an AI-enabled threat substantively (score 2). IDs are corpus record numbers (Appendix A), the citation scheme used throughout the running text; the full bibliographic entry for each ID is the `ref<ID>` key in the reference list. Vectors are non-exclusive. R69 lies outside the strict grid scope and carries unattributed authorship (Section 8). Qualitative pattern markers are in the prose above.

ID	Venue (type)	Year	Stance	AI-threat vectors
16	Energies (journal)	2026	review	adv. ML, poisoning, LLM
17	Sustain. Energy Technol. Assess. (journal)	2026	defensive	poisoning
22	Informatics and Automation (journal)	2026	bridge	adv. ML
24	Electronics (journal)	2026	review	adv. ML, poisoning
49	Sensors (journal)	2025	bridge	adv. ML
50	Energy Convers. Manag. X (journal)	2025	review	all five vectors
54	Energies (journal)	2025	defensive	adv. ML
57	Energies (journal)	2025	review	poisoning
69	Int. Res. J. Modernization (journal)	2025	bridge	adv. ML
74	Deep Blue, Univ. Michigan (thesis)	2025	defensive	poisoning
94	arXiv (preprint)	2022	review	adv. ML, poisoning

jection, biased recommendations—in the review literature (R16, R24, R50). (4) Agentic automation represented only once, in R50; no study constructs an adversarial RL or agentic pipeline. "Substantive engagement" therefore currently means backgrounding a threat while retaining a defensive detector.

5.6. Evidence quality (RQ4)

Machine learning is used in 70 of 83 full-text studies; its dominant role is detection (55), followed by forecasting (12), none (13), and control (3). Evaluation method splits as simulation 33, qualitative 28, testbed 10, conceptual 5, mixed 5, and hardware-in-the-loop 2 (R55 and R62 after adjudication). Maturity is prototype in 46, concept in 34, and validated in only 3. Baselines are reported in 41 of 83 studies and quantitative metrics in 43; both figures mix primary studies with reviews, so the non-review denominators matter and are reported with them: baselines rise to 37 of 49 non-review studies (76%) and metrics to 40 of 49 (82%), against 41/83 (49%) and 43/83 (52%)

over all full-text records. Limitations are stated in 67. Code is released in 3 of 83 studies and data in 16. The evidence base is, in short, prototyping-heavy and release-light. A venue-type filter alone (the 62 journal-classified full-text records) is not a quality filter: it retains two Zenodo deposits (R30, R32), one GSC Online Press title (R65) and one IRJMETS title (R69). A peer-review-filtered sensitivity that removes those four venues keeps 58 records and leaves the asymmetry's direction intact: 23/58 (39.7%) any, 8/58 (13.8%) substantive, 35/58 none, 55/58 tool, and the single targeted twin (R59) is the out-of-domain vehicle-twin review. Even this filter is a proxy—venue type and publisher are not a peer-review certification—and no indexing-based filter was applied; readers should weight venue quality into any figure reproduced from this review. Code and data remain low in every denominator.

Grid domains are dominated by studies that span multiple domains (*multi*, 47). Where a single domain is named: distribution 11, unclear 7, EV charging 6, microgrid 6, transmission 5, and substation 1. Among the 29 records naming a single domain, transmission (5) trails distribution (11) and EV charging (6); with 47 of 83 records coded *multi* and the field at moderate agreement ($\kappa = 0.44$), domain-level comparisons carry wide uncertainty, and we state the under-representation of transmission as an observation within this corpus rather than a claim about the literature.

5.7. Reliability results

Table 7 reports agreement on the 20-record sample (24% of the 83 full-text records). On `ai_threat_overall`, the 3×3 counts are (rows coder 1, columns coder 2): 0: 9, 1, 1; 1: 1, 3, 1; 2: 1, 2, 1. The count behind the headline, 35/83, rests on the "any engagement vs. none" collapse: coder 1 has 9 of 20, coder 2 has 9 of 20, agreement 16/20 ($\kappa = 0.60$, PABAK 0.60, bootstrap 95% CI 0.20–0.90). The three-class κ is 0.41 (PABAK 0.48); these are different statistics for different questions. On the three-level scale itself, the ordinal statistics are linear weighted $\kappa = 0.43$ (bootstrap 95% CI 0.08–0.74), quadratic weighted $\kappa = 0.44$ (0.03–0.80) and Gwet's AC1 = 0.50 (0.18–0.80): chance-corrected agreement on the ordinal scale is moderate under every statistic we computed, and the intervals are wide. The positive classes are thin: each coder labels only about 9 of 20 records as engaged and 3–4 as substantive, so at $n = 20$ neither κ nor AC1 on the substantive class can be stable. Collapsing to "substantive vs. not": agreement 15/20, but $\kappa = 0.14$ (PABAK 0.50, bootstrap 95% CI -0.23 to 0.64), because only 1

of the 6 studies either coder labelled substantive is labelled so by both (the single (2,2) cell). The two collapses are not comparably reliable.

The disagreement on "any engagement" is symmetric (2 each way), but four disagreements cannot distinguish coder bias from noise, and symmetry is not evidence of accuracy: the interval on the difference includes under- and over-counting of the same magnitude. The headline supported by the reliability data is the 35/83 any, not the 11/83 substantive, which is fragile. This reading is post hoc—formed after seeing the 3×3 table—and is flagged as such.

To quantify what the misclassification does to the headline, we propagate the sample error matrix to the corpus under the assumption that coder 2's coding is the gold standard: the coder-1-engaged records stay engaged with probability 7/9, and the coder-1-zero records flip to engaged with probability 2/11, giving an expected $35 \times 7/9 + 48 \times 2/11 \approx 36$ of 83 (43.3%; bootstrap 95% CI 21–51). The anchoring is a choice, and the choice is circular: coder 2 is a co-author of this review, so treating coder 2 as gold cannot validate the coding, and anchoring on coder 1 instead would reproduce 35/83 by construction. The interval 21–51 spans minority and majority readings of the any-level share, so it is best read not as a confidence statement but as a demonstration that the observed error structure is compatible with both—which is the honest bound on this instrument.

Fields with substantial agreement include `baseline_compared` ($\kappa = 0.90$), `twin_role` (0.77), `ai_poisoning` (0.73), `threat_model_explicit` (0.69) and `metrics_reported` (0.68)—though `twin_role`'s κ carries a bootstrap interval of 0.00–1.00 and the sample contains about 0.2 expected *target* cases, so it says almost nothing about the rare class the headline needs, and its dominant category is a near-constant default (75 of 83). Fields at moderate or fair agreement include the ones the headline most needs: `ai_threat_overall` (0.41), `ai_evasion` (0.39), and the threat-vocabulary fields, which are the least reliable data in the review at the exact-set level (agreement 5%). Re-reported at the family level—interface threats (`mitm`, `replay`, `reverse_pathway`, `false_command`), learned-model threats (`poisoning`, `evasion`, `prompt`), and classical threats (`dos`, `fdi`, `malware`, `physical`, `insider`, `supply_chain`, `hardware`, `other`)—the vocabulary improves materially: family presence/absence agrees in 17 of 20 for the interface family ($\kappa = 0.70$), 17 of 20 for the classical family (85%, $\kappa = 0.32$, prevalence-depressed), and 14 of 20 for the learned-model family ($\kappa = 0.37$; coder 2 named fewer). Table 4 and Figure 4 should therefore be read at the

family level; per-term counts remain indicative of scale and ordering only. `attack_surface_named` is the weakest field in the study on both statistics (0.14, PABAK 0.20, AC1 0.31). Where a field is near-constant, κ is undefined and we report percentage agreement instead: `stance_confidence`, `limitations_stated`, `data_available`, `ai_llm`, `ai_agentic` and `code_available`.

We report these numbers because they bound the result. The headline's core count—whether a study engages any AI-enabled threat—is coded at moderate agreement (0.60 on the collapse that produces it), so the 35/83 figure carries real uncertainty. What this design cannot deliver—an independent (non-author) coder, a sample of at least 50 enriched for engaged records, and a third-party adjudicator whose tie-break does not default to coder 1—is consolidated with the study's other design limits in Section 8 and stated there as required work rather than as achieved. The asymmetry's *direction* is supported on every denominator and scope restriction we report; its *exact magnitude* should be quoted with the interval above.

A systematic style difference between coders emerged, and the bias indices in Table 7 quantify it rather than assert it: the largest between-coder marginal differences are on `maturity` (0.35), `stance` (0.30) and `attack_surface_named` (0.30), with coder 1 assigning the default category (*high*, *yes*, *operation*, *single*) and coder 2 differentiating (*medium*, *no*, *multi*, *unclear*) and naming more attack surfaces. PABAK, which removes prevalence depression, lifts every field's reading—most sharply the near-constant ones—so where κ and PABAK disagree, the truth is that the coders agreed often but the disagreements ran one way. We record this as a codebook problem rather than a coder problem, and as a reason not to quote κ alone on this sample.

5.8. False-negative audit

The strict AI-threat lexicon flagged none of the 48 records coded as not engaging, which is consistent with the coding—but, as the methods state, the audit shares the extraction's keyword blind spot, so this "zero" is weak evidence by construction. As a positive control the same lexicon recovers only 7 of 11 known-positive studies, because some positives use generic vocabulary the lexicon lacks (e.g. R17, R22, R57, R69). Recall on score-1 (mentioned-only) engagement, which covers 24 of the 35 engaged records and is the harder case, was never measured: no score-1 positive control was run. A pattern set

cannot recognise wording it was not written for, and this audit is a pattern set applied to the output of a pattern-assisted extraction.

We therefore also ran a broader eleven-family net. Of the 48 negatives, 38 fired on some family and 23 fired on a strong family. All 23 strong firings were individually inspected and are either conventional cyberattacks discussed without any AI component or known false positives. The instructive false positives justify human coding over keyword coding: `GaN` (the semiconductor) matching "GAN" in two records (R41, R84), the author surname "Gan" in another (R90), Xenon-135 reactor physics matching "poisoning" (R34), ARP poisoning (R68), and the SolarWinds incident discussed as a data-poisoning example (R64). Two close calls were examined and cleared: R86 bypasses *classical* state-estimation bad-data detection, not an ML detector, and R15 concerns a SCADA supervisor with AI detection stated only as future work.

The 15 weak-only firings (ids 3, 7, 10, 11, 13, 18, 21, 25, 39, 45, 63, 77, 80, 88, 100) were subsequently inspected, during this revision, by an author reading the released audit windows alongside the stored abstracts—post hoc, single-reader, coded values known to the reader, with the per-record adjudication released as `R7_READBACK.md` in the repository's `analysis/` directory. Thirteen are cleared on their face: acronym lists, classical adversary prose (APT groups, MITM, FDI, Stuxnet, BlackEnergy), defensive capability claims, related-work AI mentions, twin-synthesised training data, and one future-work plan to use GANs to synthesise attack data (R11), which the codebook scores as no engagement because the plan is not a treatment. Two are borderline score-1 candidates, flagged rather than re-scored: R21's red-team RL agents (a score-1 case only under a broad reading of agentic attack automation, since the agents serve a defensive training loop) and R100's closing sentence that attackers can use deep learning to enhance attacks (the single most plausible any-level miss of the 15). If both were re-scored to 1, any-level engagement would be 37 of 83 (44.6%); no substantive (score-2) false negative was found. A seeded 12-record sample of the 48 negatives (R9, R10, R13, R19, R48, R62, R65, R79, R81, R85, R89, R93) was re-read the same way: all 12 confirmed at zero, the typical clearance being classical FDI/MITM/DoS prose, incident history, or defensive claims, and two records (R79, R89) having fired no family at all in the full-text scan. No codes were changed by either readback.

What this audit licenses is therefore narrow, and slightly wider than before the two readbacks: the false-negative channel is disclosed and now

partially inspected, not bounded. The readbacks are author-conducted and abstract-plus-window level, so they cannot replace the design that would bound the channel—an independent full-text re-read of a random sample of the 48 negatives by a coder outside the author team, with a miss-rate estimate and confidence interval propagated to the 35/83 figure. That re-read remains feasible for the authors (the full texts are in hand) and is stated here as the required next test; it cannot be reconstructed from the released keyword digests alone, because the full texts are not redistributable with the package.

5.9. Adjudication outcome

Adjudication resolved 149 field-level disagreements and changed 20 net cells from coder 1's values: `ml_family` (12), `threat_models_named` (2), `stance` (2), `eval_method` (2), `dt_type` (1) and `grid_domain` (1). Three substantive corrections came out of it: two studies move to *bridge* stance (R22, R49), and one study is a substation-domain study (R62), contrary to its initial coding. Two third-value resolutions occurred, both `eval_method` = hardware-in-the-loop (R55, R62). No headline field moved. Under the coder-1-default tie-break this is the expected outcome (Section 4), so it should not be read as evidence that the headline is robust to adjudication; the honest statement is that adjudication, in this design, is not a test of the headline at all.

6. Discussion

6.1. The asymmetry, quantified

The corpus uses machine learning in 70 of 83 full-text studies—for detection in 55—and engages AI-enabled threats in 35, substantively in 11; within the studies most obliged to model attacks on their own detectors (security-primary, non-review, detection-ML, $n = 29$) the figures are 16 any (55%, a majority) and 6 substantive (21%). The central finding is therefore *not* that attack-side engagement is absent or even rare at the any level; it is that *any-level engagement is common in this corpus while depth is not reliably measured by this instrument*: the substantive tier is minority on every denominator, on the study's weakest collapse, on a threshold that partly proxies document length, and under a misclassification interval that includes both minority and majority readings of the any-level share. Its stability rests on

the denominator and scope restrictions of Section 5.5 and on the misclassification sensitivity of Section 5.7, which re-slice the same coded data and therefore show stability rather than independent replication; the adjudication and the keyword audit can test nothing here (Sections 5.8 and 5.7). Read within this corpus—and only this corpus, retrieved by defensive vocabulary—the defensive side has built a substantial apparatus of learned detectors, while the studies building it rarely examine those detectors' own failure modes substantively. We state the finding as an asymmetry of depth, not of rate: no temporal claim is supported by a ranked, non-random corpus, and there is no external baseline for how deep engagement should be. One alternative reading cannot be excluded: that shallow engagement partly reflects the corpus's paper quality (Section 8) rather than the field—though the cited subset engages *less* than the uncited one, which softens that concern without removing it.

6.2. The twin as a target: strict count zero, loose tiers two-thirds

No grid study in this corpus treats the twin as the adversary's objective; the single strict *target* record (R59) concerns vehicle twins and falls outside the grid domain. This null is presented as a descriptive observation of the coding, not as a finding about the field, for three reasons that are now stronger than in earlier revisions. First, the codebook's strict rule makes a zero the expected result in a security setting (an asset that needs protecting is an asset an adversary may attack), and the *tool* category is a near-constant default (75 of 83). Second, the corpus's own nuclear-domain records, read during this revision, demonstrate that the strict categories are attainable and attained outside the grid frame: R91's attack scenarios against the twin's digital and communication sections (exfiltration and offline replication of a cloud twin; deception of the twin's autonomous control; 5G jamming that starves the twin) and R92's constructed hyperparameter attacks on the twin's ML models both read as *both*-role records (Section 4 on extraction). Third, the corpus contains undeveloped twin-as-object one-liners (R48, R88) that the strict rule deliberately sets aside. The honest statement is domain-scoped: within this grid-domain corpus, the twin is never the adversary's objective; in the corpus's own nuclear-domain records, it is—and the grid difference is a finding to explain, not a universal (Section 7, P1). Habeeb et al. name the twin as an expanded attack surface [6]; on the loose tiers of Table 3, 56 of 83 full-text studies name an interface or model threat bearing on the twin's data path or learned components—largely classical grid-communication at-

tacks that a twin-based defender would name in any case—and 51 of the 75 *tool* records do so while treating the twin as an asset under protection. The honest summary is therefore not that attack-surface engagement is absent, but that it is *defensive almost everywhere in the grid corpus*: the twin's surfaces are named as things to protect, and in no grid study is the twin itself the object of the attack. What we cannot speak to is the broader version of their axis—studies in which the twin's expanded surface is the analytical centre—because our coding separates objective from asset, not centre from periphery.

6.3. The lifecycle concentration, read carefully

A descriptive result with real reach, if it holds, is the lifecycle concentration: 66 of 83 studies are coded at the operation phase, one at design—and that one (R97) is outside the grid domain—none at commissioning or maintenance. Two caveats apply. First, the field carries moderate agreement ($\kappa = 0.32$, PABAK 0.59, AC1 0.54; the study's weakest field is `attack_surface_named`, $\kappa = 0.14$), and coder 1, who coded 80% of the corpus, defaults to 'operation'; the 12 studies coded 'multi' may contain design or commissioning activity that the single label suppresses. Second, the score records the phase a study *targets*, not the phases it necessarily *touches*. Read conservatively, the finding is therefore that the literature is dominated by studies of running twins, with almost no study that sets out to secure a twin before it is built, not that no design-phase security work exists. We present this as a descriptive observation on a moderate-agreement field, not as a headline result.

6.4. Why reviews are where the threats are named

The two deepest engagements are reviews (R16, R50); 5 of the 11 substantive studies (45%) are reviews against 34 of 83 (41%) in the full-text corpus, so review authorship is not over-represented, but bridge studies are (3 of 11 against 8 of 83 full-text records, 2 more in the abstract-only set). The honest reading is that reviews have the length and the licence to survey AI threats, while primary papers have neither. Document length is a clear confound at both ends: it inflates the substantive tier's accessibility for long formats—0 of 14 conference papers clear the threshold, against 9 of 62 journal articles, 1 of 4 preprints and the single thesis (Section 4)—and these numbers should not be read as evidence that "only reviewers care".

6.5. Positioning against the prior review

Relative to Habeeb et al. [6], our contribution is a grid-side probe of a narrower instance of one of their axes, with a published corpus, a closed vocabulary, and reported reliability; full scope alignment is in Section 6.2. Our closed vocabulary covers all five of their vectors (the mapping is released as a repository artifact and is not presented as evidence; Section 3), so the comparison is not competing but additive. Two caveats are structural, not rhetorical. First, their search string has no specific attack-side operator, so records enter only through security terms; the same is true of our twelve core queries, and our four adversarial-AI probes were added precisely because of that shared blind spot. The probes cannot be claimed as a completed defence, however: the released record does not establish that probe-only records entered the retained corpus (Section 4), so the shallow engagement we measure may be partly a retrieval artefact of the core vocabulary, and the asymmetry could be weaker in the un-retrieved attack-side literature than in this corpus. Resolving that requires releasing per-record query provenance and re-running the headline on the probe-only subset—the concrete step recorded in the repository's provenance manifest. Second, their corpus spans manufacturing, healthcare, water and smart city, where the digital twin's role differs from the grid's bidirectional, safety-critical loop; grid conclusions should not be generalised from it, nor it from us.

6.6. Operational reading

For operators, three interpretive notes follow (these are author readings, not coded evidence). First, a twin whose learner is trained on the same telemetry it is meant to defend has no independent ground truth and is a single point of failure that the corpus rarely names. Second, runtime detection at the operation phase is necessary, but cannot cover design-stage trust failures. Third, the near-total absence of released code and data means a twin procurement decision cannot currently be informed by reproducible attack evidence.

7. Research agenda

We state five items. Each is tied to the gap that justifies it (from Section 5), to the validation that would confirm it, and to the evidence that would *disconfirm* it, so the agenda is falsifiable rather than aspirational. One caveat applies to every disconfirmation of the form "primary studies

already do this": because the corpus is a bounded, ranked sample with a documented retrieval gap (Section 8, Search bias), such studies may be found by better retrieval as easily as by new science. Finding them would simultaneously disconfirm the gap-as-measured-in-this-corpus and narrow the distance between the corpus and the literature; the disconfirmation conditions are stated against the wider literature, not only against this corpus. Table 8 summarises.

P1 — Twin-directed attack analysis for grids.. Build the grid-domain twin-directed attack analysis that the nuclear domain already has. Justified by the tiered result (strict `target` = 0 grid studies against 56/83 naming interface or model threats in a defender frame, Table 3) *and* by the corpus's own counterexamples: R91 and R92 already do, for nuclear reactors, what no grid study in this corpus does—attack scenarios against the twin itself (R91: exfiltration and offline replication of a cloud twin; deception of the twin's autonomous control) and constructed attacks on the twin's ML models (R92: multi-rate hyperparameter attacks). The item is therefore a transfer question, not a first-contact claim: grid research should either import that analysis or state why it does not transfer. It may not transfer for real reasons—reactor twins run higher-autonomy control loops under regulator-mandated cyber-informed engineering, while grid twins face heterogeneous legacy fleets, market-coupled timescales and different trust boundaries—and articulating those conditions is part of the item. Confirmed by a reproducible attack against a grid twin in which the twin (not the plant) is the objective, with released code and a stated adaptive adversary; disconfirmed by primary grid studies that already construct such attacks (which would move the strict grid count and end this gap).

P2 — Measure lifecycle coverage, then extend it.. Extend coverage from operation to design and commissioning. Justified by operation = 66, design = 1 (out-of-domain), commissioning = 0, on a field with $\kappa = 0.32$ (PABAK 0.59, AC1 0.54)—one of the review's weaker fields, so the gap is a coding observation first and a field observation only provisionally; confirmed by twins whose trust boundaries, interface contracts and fail-safe behaviour are specified and tested before deployment, and by a double-coded re-measurement that survives the lifecycle field's reliability problem; disconfirmed by a re-measurement showing the concentration to be a coding artefact of the single-label default.

P3 — Explicit adaptive threat models for the twin's learned components.. Justified by an explicit threat model in 19/83 and an adaptive adversary in 10/83 against detection-ML use in 55/83; confirmed by studies that state the attacker's knowledge, capability and objective *relative to the twin's detectors and forecasters* (e.g. an attacker who retrains against the twin's anomaly detector or probes its feedback loop) and test a reactive attacker; disconfirmed by primary studies that already state and test such models, which the corpus currently lacks.

P4 — Defend the learned components, not just the plant.. Justified by detection-dominant ML use (55/83) against shallow poisoning and evasion engagement (19 and 9 at any level, 7 and 1 substantive); confirmed by poisoning- and evasion-robust twins under distribution shift; disconfirmed by primary studies demonstrating robust learned detectors whose robustness is evaluated against a reactive attacker.

P5 — Release the learned models, training data and attack generators.. Not just simulation scripts. Justified by code in 3/83 (3/49 non-review) and data in 16/83 against a corpus whose attack-side claims cannot be re-run by third parties; confirmed by a third party rerunning a published twin's attack evaluation unaided; disconfirmed by the corpus remaining release-light while attack-side claims multiply. This item is AI-specific in its object: what must be released are the learned models, their training data, and the attack generators evaluated against them—artifacts whose absence makes adversarial claims unverifiable in a way that releasing only simulation scripts would not fix.

Dual use and responsible disclosure.. P1 and P5, taken together, call for building twin-directed attacks and releasing attack generators. That is deliberately dual-use: an attack generator against twin ML models is an offensive capability in the wrong hands, and infrastructure-security venues are right to expect the obligations to be stated where the calls are made. Work under this agenda should follow the responsible-disclosure norms of the community: evaluate against emulators, testbeds and synthetic twins rather than operational assets; coordinate any vulnerability findings with the asset owner and, where relevant, the vendor and the relevant CERT before publication; and release attack code with usage constraints and defender-oriented documentation, so that third parties can reproduce defensive evaluations without lowering the barrier for unsophisticated attackers. These constraints are stated

here because the agenda's falsifiability depends on exactly the artifacts—attack generators, released models—that carry the dual-use risk.

P5 must be stated carefully. We measure *research attention*, not *deployment security*. A low attack-side count does not mean grids are undefended; it means the published research has not modelled these attacks. We cannot estimate either the probability of an AI-enabled attack or its expected impact from this corpus, and we do not.

8. Threats to validity

8.1. Construct validity

"AI-enabled threat" is a construct with two distinct directions lumped together: attacks *on* AI components and attacks *wielded with* AI. The coding does not separate them per record, so the headline 35/83 is an engagement figure, not an attack-type figure (Section 5.5 reports the notes-based split for the substantive studies). The false-negative audit shows a strict lexicon misses some positives (7/11 recovery), and shares the extraction's keyword blind spot, so it discloses rather than bounds the false-negative channel. We mitigated by human coding with context, by the broad eleven-family net, and by the tiered `twin_role` reporting, but the construct remains softer than the threat-vocabulary fields.

8.2. Internal validity and screening

Screening on title and abstract is imperfect, and the released record cannot quantify the loss between abstract-level and full-text reading: no per-record artifact of the abstract-level screening survives, so the gap is disclosed rather than measured. The 17 abstract-only records are listed with DOIs (Appendix A) so readers can see exactly where full-text coding was impossible.

8.3. Reliability

Agreement on the headline fields is moderate, not strong: the any-engagement collapse that produces 35/83 is $\kappa = 0.60$ (bootstrap 95% CI 0.20–0.90; the three-class κ is 0.41); `attack_surface_named` is 0.14 (PABAK 0.20). Both coders are authors of this review, the sample is 20 of 83 full-text records, and coder-1 precedence in adjudication is a declared limitation that makes adjudication uninformative about the headline. The exact magnitude of the asymmetry should therefore be read with the misclassification sensitivity (36 of 83, 95% CI 21–51), not in isolation.

8.4. External validity and coverage

This is not a PRISMA-systematic review: one database, original run dates not preserved (the only dated search artifact is the 2026-10-19 reconstructed-string snapshot of Section 4), no registered protocol, a ranked (not random) retained subset, and five codebook revisions after the protocol was 'fixed'—all of which limit how strongly the review answers its hypothesis. Because the corpus is a ranked subset, no prevalence figure here estimates the wider literature, and no sampling-theory interval (Wilson or otherwise) is reported; the misclassification sensitivity of Section 5.7 is the only interval we attach to the headline.

Seventeen of the 100 records were available only as abstracts; this is an access limitation of the authors, not a property of the literature, and those records are listed with DOIs (Appendix A) so readers can check what is missing. The 17 are not a random subset: an abstract-level reading codes 15 as *tool* and two (R91, R92, both nuclear-domain OSTI/INL reports) as twin-directed attack work; both were read during this revision (R91 in full text, R92 from its extended abstract) and both attack twin interfaces and twin ML models (Section 4), so the strict-target null carries its "grid-domain" qualifier on read evidence, not just on abstract labels. One corpus DOI does not resolve (R18, which lies within the reliability sample) and three records have no DOI; these are disclosed rather than dropped. Venue labels disagreed with Crossref in 18 of 54 cases and were corrected in favour of the registry. The broad grid term admits adjacent domains; the strict-scope sensitivity (Section 5.5) shows the headline moves by at most about one and a half percentage points under the strictest reading.

8.5. Search bias

The core grid-security vocabulary does not retrieve attack-side work; the four adversarial-AI probes were added to widen the candidate pool (approximately 2,000 records), but the released record does not establish that probe-only records entered the screened set or the final 100, and the query strings and per-record provenance were not preserved in the released package. The probes therefore provide no retrieval-bias defence: on the released evidence, the corpus derives from the core grid-security vocabulary alone—precisely the retrieval bias the probes were meant to fix—and the asymmetry may be overstated in the un-retrieved attack-side literature. The repository's provenance manifest ('SEARCH_PROVENANCE.md') states the artifacts whose release would resolve this (query strings, run dates, per-record provenance,

and a re-run of the headline on the probe-only subset); until then, the corpus should be read as a bounded sample, not a census. The three ranking components pull in opposite directions, and which dominated the realised cut is not recoverable from the released record (Section 4); the comparison that would establish it—the year distribution of the 492 eligible records against the retained 100—requires the candidate-pool export listed in the provenance manifest. What is established is the corpus's own profile: only 16 of 100 records are from 2020–2023, so foundational attack-side work from those years is the most plausibly under-represented segment in this corpus, whatever the rule that produced it. A dated snapshot with reconstructed query strings and a second-database (Crossref) run has been archived ('analysis/search_snapshot_2026-10-19/'); the original strings and per-record provenance remain the required artifacts. OpenAlex re-runs were attempted twice on 2026-10-19 (the second during this revision) and blocked both times by the shared-IP budget; re-running the search with logged queries, and ideally a second database, is a prerequisite for any attack-side claim—not future polish—and the provenance manifest records it as such.

8.6. Researcher degrees of freedom

Five codebook revisions during extraction are a real degree of freedom. They are logged, dated, and applied consistently to all 100 records, and adjudication used the same rules; but a reader who distrusts post-hoc refinement should weight the raw pre-adjudication codes, which are released frozen alongside the adjudicated matrix.

8.7. Quality appraisal, venue quality and scope

One retained record (R69) carries unattributed authorship in a low-tier venue (IRJMETS), lies outside the strict grid scope, and is simultaneously one of the eleven substantive studies and one of the four AI-attack-generation exemplars—a combination that is exactly why the venue sensitivity and DOI verification passes were necessary, and why R69 is flagged at every use: excluding it alone moves the substantive count to 10 of 82 (5 of 28 in the core subset) without changing any direction (Section 5.5). The corpus contains 67 journal articles, 17 conference papers, 5 theses, 4 preprints, 3 reports and 4 book chapters, drawn from venues of varying strictness (including IRJMETS and GSC Online Press, and two Zenodo deposits classified as journal articles); 40 of the 100 had zero citations at retrieval and 36 of 100 carry the

review stance. No formal risk-of-bias or venue-quality appraisal was performed; the venue-type filter is not a peer-review certification, which is why Section 5.6 reports a stricter peer-review-filtered sensitivity (58 records, excluding the Zenodo deposits, the GSC Online Press title and the IRJMETS title) alongside the full-corpus figures, and why readers should weight venue quality into any claim reproduced from this review. The honest consequence is that shallow engagement may partly reflect corpus quality rather than the field: the corpus is recency-heavy, citation-light and review-heavy, and while the cited subset actually engages *less* than the uncited one (Section 5.5)—which softens the concern without removing it—no available filter cleanly separates paper quality from field behaviour. The claim in the introduction is restricted to the intersection of 'digital twin' and 'cybersecurity' in power grids; substantial adversarial-ML work on grid state estimation lies outside that intersection—the false-data-injection and adversarial-ML literatures are large without ever naming a twin [2, 7, 10]—and is not covered by the headline numbers.

9. Conclusion

Read as an exploratory, corpus-level measurement, the review's defensible finding is that *any-level* AI-threat engagement in this grid digital-twin security corpus is common, while *depth* is not reliably measured by the instrument used here. Across a ranked corpus of 100 studies (a declared non-probabilistic subset of 492 eligible records, so every figure describes the corpus, not the literature; its 76% share of 2025–26 records is partly a selection artefact, and its retrieval vocabulary was defensive), machine learning is used in 70 of 83 full-text studies—for detection in 55—while 35 engage any AI-enabled threat (a majority, 16/29 and 30/55, in the core detection-ML subsets), 11 substantively—on the weakest coder collapse, under a threshold that partly proxies document length, and with a misclassification interval (21–51 of 83) that includes both minority and majority readings of the any-level share. The asymmetry the review set out to test was, in its original form, not supported; what survives is the narrower statement above, and the title reflects it. As coded—and this is a descriptive observation of the coding, not a finding about the field—no grid study treats the twin as the adversary's objective; 56 of 83 name classical interface or model threats bearing on the twin's data path or learned components, almost always as something to protect; and two read nuclear-domain records (R91, R92) that do attack twin interfaces and twin

ML models bound that null from outside the grid domain, turning P1 into a transfer question rather than a first-contact claim. Coverage, as coded, is dominated by the operation phase (66 of 83; lifecycle $\kappa = 0.32$, AC1 0.54); the single design-phase record is outside the grid domain, none are coded at commissioning or maintenance, and threat models are thin (explicit in 19/83).

These directions are robust to the denominator and scope restrictions we report (including a strict grid scope and a peer-review venue filter)—which are re-slices of the same coded data and therefore show stability, not independent replication—and to a misclassification sensitivity anchored on the second coder (36 of 83, 95% CI 21–51), whose anchoring is circular and whose interval spans minority and majority readings. They are not tested by the adjudication, whose tie-break defaults to the primary coder; the false-negative channel is now partially inspected (15 weak audit firings read; a seeded 12-record sample of the negatives re-read; no substantive false negative found), though not bounded. The closest prior review names the dual role; this review measures only the narrowest, strict-objective instance of its attack-surface axis at grid scale and finds that instance empty in the grid domain, while the looser tiers show the twin's surfaces widely engaged but almost always defensively, and the corpus's own nuclear-domain records show the strict categories attainable. The broader attack-surface axis remains untested here. The five-item agenda that follows is deliberately falsifiable: each item names the gap that justifies it, the evidence that would close it, and the evidence that would show the gap was never real—noting that "already being done" can be revealed by better retrieval as easily as by new science, and that the dual-use obligations of twin-directed attack work are stated alongside it.

Highlights

- Structured review of a ranked 100-study corpus; figures describe the corpus alone
- ML in 70/83 (detection 55); any-level engagement common, depth not reliably measured
- No grid study targets the twin itself; read nuclear records R91/R92 do

- Coverage concentrates at operation (66/83); design and commissioning near-empty
- Five-item agenda: each item tied to its gap, disconfirming evidence, dual-use limits

Declaration of competing interest

The authors declare no competing interests.

Data availability

All extraction data, the codebook, the coder sheets, the reliability script and the figure script are available in the reproducibility package accompanying this review (repository link supplied at submission). Appendix A lists all 100 included studies in the paper itself; the search-provenance manifest (`SEARCH_PROVENANCE.md`) and the dated search snapshot (`analysis/search_snapshot_2026-10-19/`) record the query-level artifacts released and those that remain outstanding, as Section 8 discloses. The round-7 readbacks (15 weak audit firings, the 12-record negatives sample, and the R91/R92 post-hoc coding) are released as `R7_READBACK.md` in the `analysis/` directory; the R91 full text is a publicly accessible INL/OSTI report cited by DOI and is not redistributed with the package.

CRediT authorship contribution statement

Author One: conceptualisation, methodology, formal analysis, writing—original draft. **Author Two:** validation, reliability coding, writing—review and editing.

Table 7: Inter-coder agreement on the blind sample ($n = 20$ of the 83 full-text records). κ is reported where interpretable; PABAK uses the codebook's category count per field; AC1 is Gwet's first-order agreement coefficient (prevalence-robust, computed on the same 3×3 or nominal tables); "bias" is the largest between-coder difference in category marginals (proportion of 20); † marks near-constant fields (modal category $\geq 90\%$ in coder 1) where κ is not interpretable and percentage agreement is reported instead. Bootstrap 95% CIs for headline fields are in the text.

Field	% agree	κ / set metric	PABAK	AC1	bias
Cohen's κ interpretable					
baseline_compared	95	0.90	0.90	0.90	0.05
twin_role	95	0.77	0.93	0.94	0.05
ai_poisoning	90	0.73	0.85	0.88	0.10
threat_model_explicit	90	0.69	0.80	0.85	0.10
metrics_reported	85	0.68	0.70	0.72	0.05
security_primary	85	0.58	0.70	0.77	0.15
ml_used	85	0.57	0.77	0.77	0.05
ml_role	70	0.55	0.64	0.64	0.10
stance	60	0.46	0.50	0.51	0.30
dt_type	75	0.46	0.69	0.72	0.15
grid_domain	65	0.44	0.61	0.61	0.15
ai_adv_ml	70	0.44	0.55	0.59	0.10
maturity	60	0.42	0.47	0.41	0.35
ai_threat_overall	**65**	**0.41**	**0.48**	**0.50**	**0.05**
ai_evasion	75	0.39	0.62	0.69	0.20
eval_method	50	0.37	0.42	0.38	0.20
lifecycle_stage	65	0.32	0.59	0.54	0.25
attack_surface_named	60	0.14	0.20	0.31	0.30
Percentage agreement only (†)					
code_available	100	†	1.00	1.00	0.00
ai_agentic	95	†	0.92	0.95	0.05
ai_llm	90	†	0.85	0.89	0.10
data_available	85	†	0.77	0.84	0.10
stance_confidence	70	†	0.55	0.60	0.20
limitations_stated	70	†	0.40	0.60	0.10
Multi-label fields (set metrics; family-level agreement in the text)					
threat_models_named	5	Jaccard 0.43	—	—	—
ml_family	20	Jaccard 0.42	—	—	—
threat_count	10	mean $\lvert\Delta\rvert$ 1.80	—	—	—

Table 8: Research agenda: gap, direction, the validation each item demands, and the evidence that would disconfirm it.

#	Gap (evidence)	Direction	Confirm / disconfirm
P1	Strict target 0 (grid); 56/83 defender-framed threats; nuclear R91/R92 counterexamples	Twin-directed attack analysis for grids	Reproducible twin-directed attack, released code / grid studies already doing so
P2	Operation 66/83 ($\kappa = 0.32$); design 1, commissioning 0	Measure, then extend, lifecycle coverage	Pre-deployment trust boundaries tested / re-measurement shows coding artefact
P3	Explicit threat model 19/83, adaptive 10/83 vs detection-ML 55/83	Adaptive threat models for learned twin components	Attacker knowledge/capability/objective stated and tested / primary studies already doing so
P4	Poisoning 19, evasion 9 (subst. 7, 1) vs detection 55	Defend learned components	Robust detectors under reactive attack / robust detectors already demonstrated
P5	Code 3/83 (3/49 non-review), data 16/83	Release learned models, training data, attack generators	Third party reruns unaided / corpus stays release-light

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Appendix A. The 100 included studies

Table 9 lists every record in the retained corpus, so that the included-study set is visible in the paper itself and not only in the released data. Citation scheme: IDs are corpus record numbers, cited in the running text as R<ID>; each record's full bibliographic entry appears in the reference list under the key `ref<ID>` (with DOI where one exists), so no separate mapping between the two schemes is needed. The IDs are a display order—year descending, and within each year block citations descending (IDs 1–45 are all 2026)—as the Cit. column shows. That ordering is a property of the export, not evidence about the selection rule that retained these 100 of the 492 eligible records; the comparison that would establish the rule is the year distribution of the 492 eligible records against the retained 100, and the candidate-pool export it requires is listed as an unreleased artifact in the search-provenance manifest (Section 4). "FT" marks the 83 records coded at full text; the 17 abstract-only records were coded at abstract level, with full-text fields treated as missing. R18, R59, R69 and R97 fall outside the strict grid scope of Section 5.5; R34, R68 and R84 are the borderline cases there. R69 carries unattributed authorship (Section 8).

Table 9: The retained corpus ($n = 100$). IDs are a display order: year descending, citations descending within each year block (Cit. column). Citation counts are as retrieved at selection. DOIs are in the reference-list entries (keys `ref<ID>`).

ID	Study (first author)	Venue (type)	Cit.	FT
1	Bo Nørregaard Jørgensen (2026)	Applied Sciences (journal)	21	Y
2	N Subramanian (2026)	Energy Reports (journal)	8	Y

Table 9 continued

ID	Study (first author)	Venue (type)	Cit.	FT
3	Muhammed Cavus (2026)	Renewable and Sustainable Energy Reviews (journal)	7	Y
4	Deepthi Pilakkat (2026)	IEEE Access (journal)	6	Y
5	Ehsan Naderi (2026)	Electronics (journal)	4	Y
6	Binastya Anggara Sekti (2026)	Advances in computational intelligence and roboti... (chapter)	4	–
7	Nagarajan Munusamy (2026)	Global Energy Interconnection (journal)	4	Y
8	Giuseppe Loseto (2026)	IEEE Sensors Journal (journal)	4	–
9	Saad Khan (2026)	IEEE Access (journal)	3	Y
10	Matteo Repetto (2026)	Applied Cybersecurity & Internet Governance (journal)	1	Y
11	Anant Upadhiyay (2026)	Discover Internet of Things (journal)	1	Y
12	Francisca Ezulike (2026)	International Conference on Cyber Warfare and Sec... (conf.)	1	Y
13	Ismail Erkek (2026)	International Journal of Information Security (journal)	1	Y
14	Syed Saad Ali (2026)	International Journal of Innovations in Science a... (journal)	0	Y
15	Roberto Ciavarella (2026)	Electricity (journal)	0	Y
16	Reham Abdullah Sanad Alsbua (2026)	Energies (journal)	0	Y
17	Hao Zhang (2026)	Sustainable Energy Technologies and Assessments (journal)	0	Y

Table 9 continued

ID	Study (first author)	Venue (type)	Cit.	FT
18	Syed Mehzeyar Ali Zaidi (2026)	Journal of Engineering and Computational Intellig... (journal)	0	Y
19	Shameem Hasan (2026)	Energy Conversion and Management X (journal)	0	Y
20	Aravind Kumar Karpoorapu (2026)	Zenodo (CERN European Organization for Nuclear Re... (chapter)	0	Y
21	Yue Wang (2026)	Systems (journal)	0	Y
22	Igor Vitalievich Kotenko (2026)	Informatics and Automation (journal)	0	Y
23	S. Rayhan Kabir (2026)	Applied Sciences (journal)	0	Y
24	Osasumwen Cedric Ogiesoba-Eguakun (2026)	Electronics (journal)	0	Y
25	Oleg V. Trubienko (2026)	Russian Technological Journal (journal)	0	Y
26	Paroma Chatterjee (2026)	West Virginia University (thesis) (thesis)	0	–
27	Kanchan Yadav (2026)	2026 IEEE 4th Int. Conf. on Power Electronics & ... (conf.)	0	–
28	Jennifer Ikeoluwa Ade-moomiyo (2026)	International Journal of Innovative Technology an... (journal)	0	Y
29	GHOREISHI, MOHAMMAD (2026)	IRIS Research product catalog (Sapienza Universit... (thesis)	0	–
30	Safa Mohamed (2026)	Zenodo (CERN European Organization for Nuclear Re... (journal)	0	Y
31	Franco Fernando Yanine (2026)	Sustainability (journal)	0	Y

Table 9 continued

ID	Study (first author)	Venue (type)	Cit.	FT
32	Subiksha GD, Anusha C, Deno Star A (2026)	Zenodo (CERN European Organization for Nuclear Re... (journal)	0	Y
33	Syed Kousar Niasi K (2026)	Preprints.org (preprint)	0	Y
34	Mehtab Alam (2026)	Frontiers in Mechanical Engineering (journal)	0	Y
35	Fahad Khan (2026)	International Journal of Advanced Research (journal)	0	Y
36	Vladyslav Pliuhin (2026)	Lighting Engineering & Power Engineering (journal)	0	Y
37	Allen Antony (2026)	Hogskolan Ihalmstad (Halmstad University) (thesis)	0	–
38	Narendra Kumar (2026)	International Journal of Digitalization (journal)	0	Y
39	Zornitsa B. Yordanova (2026)	Resilient Cities and Structures (journal)	0	Y
40	Akram AL-DHAFARI (2026)	AI and Digital Twins Under Siege (Wiley) (chapter)	0	–
41	Somayeh Soroush (2026)	Electronics (journal)	0	Y
42	Peng Xiao (2026)	Ingegneria Sismica (journal)	0	–
43	Akriti . (2026)	International Journal of Artificial Intelligence ... (journal)	0	Y
44	Kabiru Alani Hassan (2026)	Journal of Multi-Disciplinary Engineering Reviews... (journal)	0	Y
45	Emmanuel Emenike Ezeoba (2026)	Advances in Machine Learning & Artificial Intell... (journal)	0	Y

Table 9 continued

ID	Study (first author)	Venue (type)	Cit.	FT
46	T. A. Rajaperumal (2025)	Energy Informatics (journal)	118	Y
47	Sohel Rana (2025)	American Journal of Advanced Technology and Engin... (journal)	87	Y
48	Herman Zahid (2025)	Results in Engineering (journal)	30	Y
49	Ali Sayghe (2025)	Sensors (journal)	29	Y
50	Yaser Mike Banad (2025)	Energy Conversion and Management X (journal)	23	Y
51	Hind Lakhal (2025)	IEEE Access (journal)	22	Y
52	Muhammed Cavus (2025)	Energies (journal)	19	Y
53	Reza Hemmati (2025)	Energy Conversion and Management (journal)	18	–
54	Bo Li (2025)	Energies (journal)	15	Y
55	Samuel Abiodun Oyedotun (2025)	Journal of Science Research and Reviews (journal)	14	Y
56	Kwabena Addo (2025)	Energies (journal)	11	Y
57	Ernest Fiko Morgan (2025)	Energies (journal)	10	Y
58	Nabeel A. Y. Al-Qirim (2025)	International Journal of Engineering Business Man... (journal)	6	Y
59	Uzair Muzamil Shah (2025)	Smart Cities (journal)	5	Y
60	Pradeep Chintale (2025)	Auerbach Publications eBooks (chapter)	3	–
61	Evgenii Sergeevich Mityakov (2025)	Computational nanotechnology (journal)	2	–
62	Mansi Girdhar (2025)	IEEE Access (journal)	2	Y
63	He Jian Wu (2025)	Preprints.org (preprint)	2	Y
64	Ghiulnar - Yelda Cherim (2025)	International Symposium on Advanced Topics in Ele... (conf.)	1	Y

Table 9 continued

ID	Study (first author)	Venue (type)	Cit.	FT
65	Pedro Barros (2025)	World Journal of Advanced Research and Reviews (journal)	1	Y
66	Muhammad Arbab Khan (2025)	Global Power, Energy and Communication Conference (conf.)	1	Y
67	Samir Thapa (2025)	IEEE International Conference on Electro Informat... (conf.)	1	Y
68	Vita Santa Barletta (2025)	Biannual Conference of the Italian SIGCHI Chapter... (conf.)	1	Y
69	Unattributed (2025)	International Research Journal of Modernization i... (journal)	0	Y
70	Li, Jin (2025)	Espace ETS (ETS) (report)	0	Y
71	Saif Ali (2025)	Euro vantage journal of artificial intelligence (journal)	0	Y
72	Dalila Anseri (2025)	2025 4th Int. Conf. on Theoretical and Applicativ... (conf.)	0	–
73	Bilal Khan (2025)	International Conference on Electrical, Communica... (conf.)	0	Y
74	Mansi Girdhar (2025)	Deep Blue (University of Michigan) (thesis)	0	Y
75	Artem I. Zemtsov (2025)	Gylym zane bilim (journal)	0	Y
76	Abraham Ayodeji Abayomi (2025)	International Journal of Scientific Research in C... (journal)	0	Y

Table 9 continued

ID	Study (first author)	Venue (type)	Cit.	FT
77	Opy Das (2024)	Energy Conversion and Management X (journal)	107	Y
78	Anurag Kumar Srivastava (2024)	IEEE Power and Energy Magazine (journal)	23	–
79	Nabeel A. Y. Al-Qirim (2024)	IEEE Open Journal of the Communications Society (journal)	12	Y
80	Baza Somda (2024)	International Conference on Intelligent Systems A... (conf.)	7	Y
81	Timotius Oentung (2024)	Resilience Week (RWS) (conf.)	5	Y
82	Tommaso Zoppi (2024)	IEEE Pacific Rim International Symposium on Depen... (conf.)	3	Y
83	Abdulmueen Alrashide (2024)	Proceedings of the Annual Conference of the IEEE ... (conf.)	1	Y
84	Masoud Amirrezai (2024)	IEEE Industrial Electronics Society Annual On-Lin... (conf.)	1	Y
85	Prabhat Kumar (2023)	Solar Energy (journal)	101	Y
86	Mansoor Ali (2023)	IEEE Transactions on Information Forensics and Se... (journal)	91	Y
87	Erdal Irmak (2023)	Energies (journal)	67	Y
88	Omer Sen (2023)	IEEE International Conference on Communications, ... (conf.)	13	Y
89	Gholamreza Nasiri (2023)	Machines (journal)	12	Y
90	Luka V. Strezoski (2023)	Energija Ekonomija Ekologija (journal)	4	Y

Table 9 continued

ID	Study (first author)	Venue (type)	Cit.	FT
91	Christopher M. Spirito (2023)	Idaho National Laboratory / OSTI (report) (report)	1	–
92	Christopher M. Spirito (2023)	Idaho National Laboratory / OSTI (report) (report)	1	–
93	Md. Mhamud Hussen Sifat (2022)	Energy and AI (journal)	174	Y
94	Tianming Zheng (2022)	arXiv (Cornell University) (preprint)	12	Y
95	William Danilczyk (2021)	North American Power Symposium (NAPS) (conf.)	79	–
96	Amardeep Balasaheb Shitole (2021)	IEEE International Conference on Smart Technologi... (conf.)	12	Y
97	Yun Guo (2021)	International Conference on Power System Technolo... (conf.)	11	Y
98	Nandha Kumar Kandasamy (2021)	arXiv (Cornell University) (preprint)	9	Y
99	Ahmed Aly Saad Ahmed (2021)	Florida International University (thesis) (thesis)	2	–
100	Ahmed Saad (2020)	IEEE Transactions on Smart Grid (journal)	238	Y